

Business Report: Geldium Delinquency Risk Strategy

Prepared for: Head of Collections, Geldium Financial Services

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Step 1 Summary of Predictive Insights

Key Customer Segments at High Delinquency Risk

Analysis of the Geldium dataset reveals three distinct high-risk customer segments: (1) **Over-leveraged borrowers** — customers with credit utilization exceeding 80%, indicating near-maximum debt capacity; (2) **Chronic late payers** — customers with two or more missed or delayed payments in recent months, representing the strongest historical signal of future default; and (3) **Low-income, high-DTI customers** — individuals whose Debt-to-Income ratio exceeds 40%, leaving minimal financial buffer for unexpected expenses. Notably, high-income customers with aggressive spending patterns also appear in the high-risk tier due to 'lifestyle creep' behaviors.

Variables Most Strongly Influencing Delinquency

Based on the Random Forest feature importance analysis, the following variables are the primary drivers of delinquency prediction:

Ran k	Variable	Why It Matters
#1	Credit Utilization Rate	Over-reliance on available credit; strongest signal of financial over-extension.
#2	Payment History	Missed or delayed payments are the most direct predictor of future default behavior.
#3	Debt-to-Income (DTI) Ratio	High DTI indicates limited repayment capacity and financial stress.
#4	Annual Income	Low income constrains ability to service debt, especially during disruptions.
#5	Open Credit Lines	Multiple open lines suggest broad credit exposure and potential overcommitment.

Top 3 Risk Factors (Quick Reference)

- **High Credit Utilization (>80%)** — Primary over-extension indicator; triggers immediate risk flag.
- **Missed / Delayed Payments (2+)** — Strongest historical predictor; signals intent and capacity issues.
- **High DTI Ratio (>40%)** — Indicates insufficient financial headroom for loan servicing.

Step 2 Recommendation Framework

SMART Business Recommendation

S — Specific	Introduce a proactive outreach program targeting customers with credit utilization above 75% and one or more missed payments, offering personalised repayment plans or credit counselling.
M — Measurable	Reduce the 90-day delinquency rate in the high-risk segment by 15% within 6 months , tracked via monthly cohort reporting.
A — Achievable	Leverage the existing collections team and AI model output (probability scores) to prioritise the top 20% highest-risk accounts for immediate outreach.

R — Relevant	Directly addresses Geldium's core objective of reducing financial loss from delinquent accounts while improving customer retention.
T — Time-bound	Pilot programme to launch within 30 days of model deployment; full evaluation at the 6-month mark.

Stakeholder Rationale: By identifying at-risk customers *before* they miss a second payment, Geldium can intervene early — reducing write-offs, preserving customer relationships, and demonstrating responsible lending practices to regulators.

Step 3 Ethical & Responsible AI Considerations

Fairness Risks and Mitigation Strategies

Fairness Risk	Description	Mitigation Strategy
Risk 1: Demographic Bias	If location or age fields correlate with protected characteristics, the model may unfairly flag certain groups as high-risk — even without intent.	Apply Disparate Impact Ratio (DIR) checks. Remove or anonymise demographic proxies. Audit model outputs by subgroup regularly.
Risk 2: Historical Data Bias	Training data may reflect historical lending biases — over-representing defaults in certain income bands due to previous unequal access to credit.	Use re-sampling techniques (SMOTE) to balance the training set. Validate model performance equally across all income groups.

Explainability for Non-Technical Stakeholders

Rather than presenting raw probability scores, collections staff will receive a simplified **Risk Traffic Light** system: **Red** (high risk — immediate action), **Amber** (moderate — monitor closely), **Green** (low risk — standard process). Each flag will include a plain-language reason such as: *"This customer has been flagged because their credit card balance is near the limit and they missed a payment last month."* This approach ensures decisions are transparent, human-reviewed, and explainable to customers if challenged.

Responsible AI Commitment: All model predictions will be treated as decision-support tools, not automated decisions. A human collections officer will review every high-risk flag before any customer action is taken. Model performance will be audited quarterly for drift, bias, and accuracy.